

Pokémon TCG Marketplace and Collection Management Platform

Initial Course Project Statement

Student Project Brief

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Context

The Pokémon Trading Card Game (TCG) has a diverse ecosystem of collectors, players, independent sellers, specialized businesses, and online communities. In Colombia, information and transactions related to the hobby are distributed across social networks, messaging applications, specialized stores, and international price-tracking platforms. This fragmentation makes it difficult for users to discover products, compare prices, evaluate sellers, track their collections, and obtain a consistent view of the state of the hobby.

The project proposes a platform that connects individual users with Pokémon TCG businesses and sellers while combining marketplace capabilities with a centralized source of information about cards, products, prices, collections, and availability. Rather than treating the platform exclusively as an online store, the system should provide a broader environment in which users can explore the Pokémon TCG ecosystem, manage their collections, discover cards and products, evaluate purchasing opportunities, and interact with sellers.

Problem Statement

Pokémon TCG Colombian users currently need to rely on multiple platforms to perform activities that are closely related. A collector may use one service to research cards and sets, another to estimate prices, social networks to find sellers, messaging applications to negotiate purchases, and separate tools to organize their collection. These fragmented experiences make it difficult to establish a consistent relationship between cards, market prices, listings, sellers, and a user's personal collection.

Sellers and businesses also lack a specialized environment that combines product listings with relevant information about the Pokémon TCG market. Businesses may require greater visibility and credibility, while individual sellers may prefer a lightweight way of advertising products and communicating directly with potential buyers without necessarily processing the transaction through the platform.

The project therefore requires a platform that can connect the information and interaction layers of the hobby: a structured catalog of Pokémon TCG products and cards, marketplace listings, seller profiles and reputation, price history, collection management, and mechanisms for discovering and planning future acquisitions.

Project Intention

The purpose of this project is to specify and later design a platform that centralizes discovery, commerce, market information, and collection management for the Pokémon TCG ecosystem in Colombia.

The system should allow users to explore a structured catalog of cards and sealed products, discover available listings, understand historical and recent prices, interact with verified businesses and individual sellers, and maintain one or more personal collections. Users should also be able to organize cards according to their own collecting interests, visualize their collections through virtual binders, and maintain lists of cards they want to acquire.

At this initial stage, the project should be approached as a whole-system specification challenge rather than as a predetermined set of technical modules. The first objective is to understand the different actors, their goals, relationships, business rules, constraints, and interactions before determining how the platform should later be decomposed or implemented.

Stakeholders

The project involves multiple stakeholders with different goals and constraints:

- **Collectors and hobbyists** need reliable information about cards, prices, availability, and sellers, as well as tools to organize and track their collections and plan future acquisitions.
- **Buyers and casual users** need to discover Pokémon TCG products, compare available listings, understand their market context, and communicate with sellers through a convenient process.
- **Individual sellers** need a simple way to publish cards, bundles, and products, expose their listings to interested users, and establish direct communication with potential buyers without necessarily processing transactions through the platform.
- **Pokémon TCG businesses and specialized stores** need greater visibility, a professional profile, a mechanism for publishing their inventory, and a way to establish credibility through verified business information and user reviews.
- **Platform administrators** need to review and approve business applications, moderate listings and user-generated information, manage platform policies, and maintain the reliability and integrity of the marketplace and catalog.
- **The Pokémon TCG community** benefits from a centralized source of structured information about cards, sets, prices, products, and collecting trends, even when the information is not directly associated with a transaction.

High-Level Goals

The system is expected to support the following goals:

- Provide a centralized and structured catalog of Pokémon TCG cards and products.
- Allow users to discover and filter cards according to characteristics such as sets, eras, Pokémon, colors, styles, artists, and other collecting criteria.
- Connect cards and products with available marketplace listings when such listings exist.
- Provide price information, recent sales, and historical price trends to give users market context.
- Allow businesses and individual sellers to publish cards, card bundles, and sealed Pokémon TCG products.
- Distinguish between verified businesses and individual sellers, applying different registration and transaction rules to each type of seller.
- Allow users to communicate with individual sellers through external channels (such as any messaging app), including relevant product and price information in the generated contact message.
- Provide user profiles, business profiles, and reputation mechanisms such as reviews.
- Allow users to add cards and products to one or more personal collections regardless of whether they were acquired through the platform.
- Track the value and historical evolution of users' collections.
- Provide virtual binder views that allow users to organize their collections according to their own preferences.
- Allow users to create different collection groupings, such as general collections, sealed products, cards, cards intended to be kept, or cards intended for future sale.
- Distinguish between products acquired through the platform and products added manually by the user.
- Allow users to maintain wish lists or acquisition-planning lists containing cards they are interested in obtaining.
- Support acquisition planning based on factors such as price, availability, and seller or product location.

Operational Context

The platform should operate over a structured Pokémon TCG catalog containing individual cards as well as sealed products and card bundles. A card in the catalog does not necessarily represent an item currently available for sale. Instead, it represents an entity in the broader Pokémon TCG ecosystem that may have associated marketplace listings, price information, collection references, or other information.

Marketplace listings represent specific offers made by businesses or individual sellers. Businesses would register through an application process requiring relevant legal and business information, such as their legal identity and external presence through an Instagram profile or website. These applications would remain pending until reviewed and approved by a platform administrator.

Individual sellers would have a different interaction model. They could publish listings without becoming verified businesses, but transactions would not necessarily be processed directly by the platform. Instead, interested users could initiate contact through an external messaging app, with the platform generating a predefined message containing information about the selected product and its listed price.

Card bundles require additional internal tracking because their composition affects the information maintained by the platform. When creating a bundle, a seller should identify the individual cards that make up the bundle so that the system can associate the bundle with the corresponding catalog entities and account for those cards appropriately.

Users may acquire products outside the platform and still register them in their collections. Conversely, when a user purchases a product through the platform, the system should be able to offer the user the option of adding that product to a collection. This creates a distinction between the marketplace history of a product and its presence in a user's personal collection.

Constraints and Tensions

The specification should take into account that the platform combines several different responsibilities that may conflict with one another.

The catalog must distinguish between general information about a Pokémon TCG card and an actual product available for sale. Not every card should have a listing, and the existence or absence of a listing should not determine whether the card belongs in the catalog.

Price information also requires careful interpretation. A displayed price may represent a current listing, a recent transaction, or a historical market value, and these concepts should not be treated as interchangeable. The system therefore needs to maintain clear distinctions between market information and seller-provided prices.

The platform must also balance the openness of individual selling with the trust requirements of a marketplace. Individual sellers should be able to publish products without undergoing the same registration process as businesses, while businesses require additional verification before being represented as approved commercial entities.

Collection management introduces another distinction: ownership is not necessarily equivalent to purchasing through the platform. Users may manually add cards they already own, organize them into different collections, or classify them according to their own intentions, including cards they want to keep or cards they may eventually sell.

The system must also support multiple ways of organizing information without imposing a single collecting philosophy. A user may want to organize cards by acquisition date or value, while another may primarily care about sets, Pokémon, artists, colors, or binder layouts.

Finally, the platform must establish clear boundaries around user-generated content, seller information, reviews, listings, and market data so that the reliability and consistency of the information presented to users can be maintained.

Scope of the First Specification

For the first SDD iteration, the focus is on producing a system-level specification of the whole problem.

This includes identifying the actors involved in the Pokémon TCG ecosystem, their goals and relationships, the distinction between catalog entities and marketplace listings, the rules governing businesses and individual sellers, the main capabilities required for marketplace interactions and collection management, the relationships between cards, bundles, products, prices, and collections, and the constraints surrounding user-generated and market information.

The specification should also clarify the different acquisition scenarios, including products purchased through the platform, products acquired elsewhere, and products that users intend to acquire in the future.

At this stage, the work should not prematurely lock the architecture, database design, internal module decomposition, or implementation technologies. The emphasis should remain on clarifying what the platform needs to accomplish and under what conditions before determining how the solution will be organized internally.

Out of Scope for the First Iteration

The following should not be treated as fixed in the initial specification:

- The final internal module split of the platform.
- The exact database schema and data model.
- The exact user interface and visual design.
- The final deployment architecture.
- The exact technology stack.
- The definitive implementation of external integrations such as the external messaging app, Instagram, or external Pokémon TCG data sources.
- The exact algorithms used for price estimation or market-value calculation.
- The final business model, commissions, or monetization strategy.
- The detailed logistics and physical delivery process for marketplace transactions.

These elements may emerge later during planning and design, but they should not replace the initial work of clarifying the problem space, stakeholder expectations, business rules, and system boundaries.

Expected First Deliverable

The first deliverable should be a clear, structured specification of the system from the stakeholder perspective. It should describe:

- The problem faced by Pokémon TCG users, sellers, and businesses in Colombia.

- The relevant stakeholders and their goals.
- The distinction between the Pokémon TCG catalog, marketplace listings, market information, and personal collections.
- The main capabilities the system must provide.
- The rules governing business verification and individual sellers.
- The different ways users can acquire and register products in their collections.
- The principal constraints, rules, and tensions surrounding marketplace interactions, price information, collection ownership, and user-generated content.
- The assumptions the team is making about the Pokémon TCG ecosystem and its users.
- The open questions that still require clarification.
- The boundaries of what the platform is and is not responsible for.

A strong first specification should reduce ambiguity about the problem space without prematurely freezing implementation details. The objective is to establish a shared understanding of how a Pokémon TCG marketplace, information platform, and collection-management system should behave before planning and construction become the dominant concerns.

Starting Question for the Team

How should a platform connect Pokémon TCG buyers, collectors, businesses, and individual sellers in Colombia while providing reliable market information, flexible collection management, and a coherent environment for discovering and planning acquisitions?